

Privacy-Preserving Explainable AI via Explanation Masking against Record Reconstruction Attacks for Healthcare

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Abstract

Machine learning models have been used in many areas recently, including healthcare. However, these models are almost black-box, which have complex internal structures and make it difficult to ensure fairness and transparency. Therefore, explainable AI (XAI) has gained attention as a technique to explain the behavior of black-box models. Many Machine-Learning-as-a-Service (MLaaS) platforms provide XAI techniques such as SHAP and LIME to explain the relationship between input features and model outputs. On the other hand, in 2022, Luo et al. showed the reconstruction risk of the hidden input features from the model explanations based on Shapley values, highlighting a privacy risk in releasing explanations (e.g., leakage of sensitive patient attributes). However, it is still unclear how the reconstruction risk changes when the quality of explanations (explainability) changes, and how such risk can be mitigated for privacy-preserving AI. Therefore, in this work, we evaluate how the record reconstruction risk changes when the explainability of SHAP and LIME changes by masking explanation vectors. We conduct experiments using open datasets and show that the record reconstruction risk increases when the explainability is high. These results suggest that explanation masking can be an effective approach toward privacy-preserving explainable AI against record reconstruction attacks.

Keywords: XAI, explainability, record reconstruction attack, SHAP, LIME, healthcare, privacy-preserving AI

1 Introduction

Machine Learning (ML) models have been used in various important use cases recently, such as finance [1], healthcare [2], and e-commerce [3]. In particular, some types of models such as deep neural networks and ensemble models have complex internal structures that are “black boxes,” making it difficult to analyze their behavior internally and understand how the models arrive at decisions. Therefore, explainable AI (XAI) techniques have gained attention to ensure model transparency and explain the relationship between input features and model outputs [4, 5].

On the other hand, it is known that XAI values can leak private input features. For instance, in 2022, Luo et al. [6] showed that it is possible to estimate the hidden input records from the

model explanations using Shapley values. Practitioners increasingly expose explanations; however, releasing them may expose features of the underlying inputs (privacy leakage). This issue is particularly salient in healthcare applications, where explanations may be shared with clinicians, patients, auditors, or external service providers to justify model-based decisions [2]. For example, when models are trained on healthcare-derived tabular records (e.g., insurance claims/receipt data or electronic health records), local feature attributions can unintentionally reveal sensitive patient attributes. Therefore, a practical mechanism is needed to disclose explanations while controlling the privacy-utility trade-off in a manner suitable for healthcare settings. This hinders deploying XAI in privacy-preserving AI settings. We need a rigorous, reproducible study of this leakage and practical mitigation policies with quantified privacy-utility trade-offs.

However, it is still unclear how the reconstruction risk changes when the explainability of XAI decreases. Therefore, in this work, we evaluate how the record reconstruction risk changes when the explainability of SHAP [7] and LIME [8] changes. Especially, we change the explainability of each model explanation by masking the explanation vector. Moreover, we propose a new masking method to hide some parts of explanation vector, Privacy-aware Masking, shown in Algorithm 1. Our proposed method can control the trade-off between privacy and utility, and outperform the naive masking based on MoRF/LeRF and the random masking.

Additionally, we evaluate the decrease in explainability caused by masking model explanations using infidelity [9] and clarify the trade-off between explainability and record reconstruction risk. We report the results of applying the proposed method to three open datasets. Also, we compare our proposed method with the naive method and the random baseline. Furthermore, we discuss the limitations of our proposal and the ethical considerations.

The main contributions of current work are:

- *Threat-modeled reconstruction from explanations.* We formalize record reconstruction attacks where an adversary sees only an explanation vector of a target record and trains a model to map explanations to inputs.
- *Privacy-utility trade-offs via masking.* We evaluate masking policies that hide a fraction of explanation coordinates and quantify; (i) attacker success (MAE and Success Rate) and (ii) remaining explanation faithfulness (e.g., infidelity).
- *Empirical Findings.* Increasing the masked rate consistently reduces reconstruction success, while standard explanation infidelity correlates weakly with privacy risk.
- *A simple defense.* We propose and evaluate a *privacy-aware masking* policy that outperforms naive MoRF/LeRF and random at matched usefulness.

2 Related Work

2.1 XAI Techniques

XAI methods are broadly classified into global and local approaches. Global methods estimate dataset-level feature importance, such as QII [10] and SAGE [11], but they do not explain individual predictions. In contrast, local methods produce per-record attributions; LIME [8] fits a simple surrogate model around a target input, and Shapley-based explainers such as SHAP [7, 12] assign credit to features using Shapley values. Many other local explainers exist (e.g., DeepLIFT [13], Grad-CAM [14], Anchors [15], and MLFEM [16]), and major MLaaS platforms provide such explanations in practice [17–19]. As an application of XAI techniques, Davoodi et al. [20] proposed a post-hoc explanation method for reinforcement learning that explains safety-related risks associated with agent failures using feature-based state-transition models.

In this work, we focus on SHAP and LIME because they are widely deployed and directly match our threat model of an attacker observing a local attribution vector.

2.1.1 Privacy Risks of XAI

Releasing explanations can amplify privacy leakage beyond standard prediction APIs. Prior work reports several threats, including membership inference [21–23], model extraction [21, 24–26], and attribute/feature inference [6, 27]. Liu et al. [23] showed that explanations (including SHAP and LIME) can increase membership inference success, and Yan et al. [25] demonstrated that

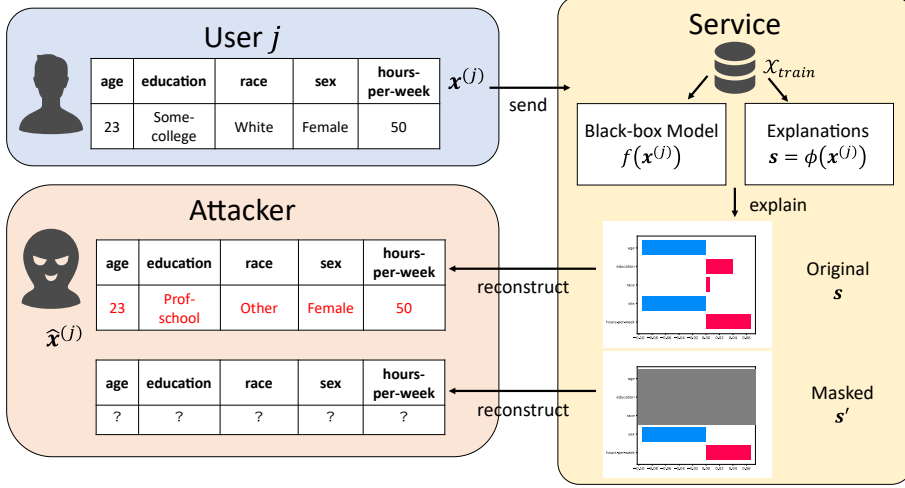


Fig. 1: Overview of record reconstruction attack from masked explanations.

explanations may facilitate model extraction. Most closely related to our study, Luo et al. [6] demonstrated record/feature inference from Shapley-value-based explanations under an auxiliary-data assumption.

However, how reconstruction risk changes when the disclosed explanation is intentionally degraded (e.g., by masking a subset of coordinates) is still not well understood, especially across different explainers under explicit utility constraints.

2.1.2 Explainability Metrics

Explainability metrics span multiple categories [28]. Since our goal is to study privacy-utility trade-offs under explanation masking, we focus on faithfulness-oriented metrics that relate attributions to model behavior under perturbations. MoRF/LeRF [29] evaluates how predictions change when features are removed in a relevance order, while infidelity [9] quantifies the mismatch between attribution-implied and actual output changes under perturbations. Related metrics include correlation-based faithfulness [30] and variants such as F-Fidelity [31]; ground-truth explanation datasets have also been used for evaluation [32].

In this paper, we use infidelity as the primary faithfulness metric and complement it with perturbation-based usefulness measures derived from MoRF/LeRF (defined in the Preliminaries) to connect explanation quality to privacy risk.

3 Problem Statement

3.1 Threat Model

Let n be the number of input features and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a black-box classification model. For a target input $\mathbf{x} \in \mathbb{R}^n$, attacker observes a local explanation vector $\mathbf{s} = \phi(\mathbf{x})$ (e.g., SHAP/LIME attributions). Optionally, attacker also observes the model score $f(\mathbf{x})$ if the API reveals it. The attacker owns an *auxiliary dataset* X_{aux} (size $|X_{aux}|$) of inputs for which it can obtain explanations to train an attack model. The attacker trains an *attack model* $\psi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ that maps explanations to inputs using pairs $\{(\phi(\mathbf{u}), \mathbf{u}) : \mathbf{u} \in X_{aux}\}$. The defender controls a *masking policy* that hides a fraction of coordinates in $\mathbf{s}$ before disclosure: masked explanation $\mathbf{s}'$. Given $\mathbf{s}'$, the attacker applies the attack model ψ to reconstruct the input: $\hat{\mathbf{x}} = \psi(\mathbf{s}')$. Figure 1 illustrates the record reconstruction attack from masked explanations.

3.2 Metrics

To evaluate the record reconstruction risk, we use two metrics: the attacker’s Mean Absolute Error (MAE) and Success Rate (SR). To evaluate the utility of XAI, we use the infidelity [9]. In addition,

we define the Masked Rate (MR) to represent the ratio of masked coordinates in the explanation vector.

Attacker’s MAE. *Mean Absolute Error* (MAE) is the average of a set of absolute errors. The *Attacker’s MAE* of the estimated vector $\hat{\mathbf{x}}$ for the vector $\mathbf{x}$ is: $MAE(\hat{\mathbf{x}}, \mathbf{x}) = \frac{1}{n} \sum_{i=1}^n |\hat{x}_i - x_i|$.

Success Rate. *Success Rate* (SR) is the fraction of features reconstructed correctly. For a numeric feature i , we count it as correct if $|\hat{x}_i - x_i| \leq \epsilon_i$ where $\epsilon_i = \alpha \cdot (\max_{\mathbf{x} \in X_{\text{aux}}}(x_i) - \min_{\mathbf{x} \in X_{\text{aux}}}(x_i))$ and $\alpha = 0.05$. For a categorical feature i encoded by a one-hot group $G_i \subseteq \{1, \dots, n\}$, we decode $\hat{c}_i = \arg \max_{j \in G_i} \hat{x}_j$ and $c_i = \arg \max_{j \in G_i} x_j$, and it is correct iff $\hat{c}_i = c_i$. Thus, $SR(\hat{\mathbf{x}}, \mathbf{x}) = (\# \text{ correct features})/n$.

Infidelity [9]. Given M random masks $z_1, \dots, z_M \in \{0, 1\}^n$, a black-box model f , an explanation function ϕ , and an input vector $\mathbf{x}$, the *infidelity* is: $INFD(\phi, f, \mathbf{x}) = \frac{1}{M} \sum_{m=1}^M (\Delta f_m - \Delta e_m)^2$, where $\Delta f_m = f(\mathbf{x}) - f(\mathbf{x} \odot (1 - z_m) + \mathbf{x}^{(0)} \odot z_m)$ and $\Delta e_m = \langle \phi(\mathbf{x}), z_m \odot (\mathbf{x} - \mathbf{x}^{(0)}) \rangle$. In this work, we set $\mathbf{x}^{(0)} = \mathbf{0}$ and $M = 100$.

Usefulness (Deletion/Comprehensiveness). Given the black-box model f , the vector $\mathbf{x}$ and the masked vector $\mathbf{x}^{(k)}$ by masking k features from $\mathbf{x}$, we define two usefulness metrics: *Deletion@k* = $|f(\mathbf{x}) - f(\mathbf{x}^{(k)})|$, *Comprehensiveness@k* = $|f(\mathbf{x}^{(k)}) - f(\mathbf{x}^{(n)})|$.

Masked Rate. Given the masked explanation vector $\mathbf{s}' \in \mathbb{R}^n$, we define the *Masked Rate* (MR) as: $MR(\mathbf{s}') = \frac{k}{n}$, where k is the number of masked elements in $\mathbf{s}'$.

3.3 Masking Problem

Our goal is to evaluate how the record reconstruction risk changes when the explainability of XAI decreases. To this end, we consider masking policies that hide a fraction of explanation coordinates before disclosure, and evaluate the trade-off between the record reconstruction risk (attacker’s MAE and SR) and the utility of model explanations (infidelity and Deletion/Comprehensiveness).

Given an explanation vector $\mathbf{s} \in \mathbb{R}^n$ for an input record $\mathbf{x} \in \mathbb{R}^n$, we determine a set of indices $I \subseteq [n]$ to mask before disclosure so that: (i) the masked explanation leaks as little information as possible to an attacker, and (ii) the explanation remains useful/faithful.

For any n -dimensional vector $\mathbf{x} \in \mathbb{R}^n$ and index $j \in [n]$, we define the masking function: $(\text{mask}(\mathbf{x}, j; \mathbf{x}^{(0)}))_i = x_i^{(0)}$ if $i = j$, and x_i otherwise, where $\mathbf{x}^{(0)}$ is a constant vector used for masking, including the zero vector and the mean vector.

3.4 Masking Policies

We consider the following masking policies to hide a fraction of explanation coordinates before disclosure.

Global MoRF/LeRF. Build one ranking $\mathcal{O}$ (e.g., by average $|\mathbf{s}|$); *MoRF*: mask top- k ; *LeRF*: mask bottom- k .

Per-sample MoRF/LeRF. For each record, build its own ranking from $|\mathbf{s}|$; mask top/bottom k .

Random. Mask k coordinates uniformly at random (baseline).

Privacy-aware masking (proposed). We propose a new masking method, Privacy-aware Masking, which aims to balance privacy and utility, showing in Algorithm 1. We use the attacker’s MAE as the privacy metric $\mathcal{P}(\mathbf{x}, \mathbf{s})$ and the Deletion as the usefulness metric $\mathcal{U}(\mathbf{x}, \mathbf{x}')$. However, other metrics can also be used. $\mathcal{P}(\mathbf{x}, \mathbf{s})$ indicates how accurate the reconstruction from the explanation $\mathbf{s}$ is to the original input record $\mathbf{x}$, and $\mathcal{U}(\mathbf{x}, \mathbf{x}')$ indicates how far the model output $f(\mathbf{x}')$ for the target class is from the original model output $f(\mathbf{x})$. In addition, ε is a small constant to avoid division by zero. The privacy metric $\mathcal{P}$ should be maximized, and the usefulness metric $\mathcal{U}$ should be minimized. Therefore, we select the index j^* that maximizes the ratio of the increase in privacy to the decrease in usefulness at each step.

Algorithm 1: Privacy-aware Masking

Input: input record $\mathbf{x} \in \mathbb{R}^n$, explanation vector $\mathbf{s} \in \mathbb{R}^n$, number of masks $k \leq n$, privacy metric $\mathcal{P}$, usefulness metric $\mathcal{U}$

- 1 $I^{(0)} \leftarrow \emptyset; \mathbf{x}^{(0)} \leftarrow \mathbf{x}; \mathbf{s}^{(0)} \leftarrow \mathbf{s};$
- 2 **for** $t = 1, \dots, k$ **do**
- 3 $j^* \leftarrow \arg \max_{j \in [n] \setminus I^{(t-1)}} \frac{\mathcal{P}(\mathbf{x}, \text{mask}(\mathbf{s}^{(t-1)}, j)) - \mathcal{P}(\mathbf{x}, \mathbf{s}^{(t-1)})}{\mathcal{U}(\mathbf{x}, \text{mask}(\mathbf{x}^{(t-1)}, j)) - \mathcal{U}(\mathbf{x}, \mathbf{x}^{(t-1)}) + \varepsilon};$
- 4 $I^{(t)} \leftarrow I^{(t-1)} \cup \{j^*\}; \mathbf{x}^{(t)} \leftarrow \text{mask}(\mathbf{x}^{(t-1)}, j^*); \mathbf{s}^{(t)} \leftarrow \text{mask}(\mathbf{s}^{(t-1)}, j^*);$
- 5 **return** masked index set $I^{(k)}$ and $\mathbf{s}^{(k)}$.

4 Experimental Settings

4.1 Datasets

The experiments use three binary-class tabular datasets: Adult (48,842 records, 14 features, 2 target classes) [33], Bank Marketing (45,211 records, 16 features, 2 target classes) [34], and Credit Card Client (30,000 records, 24 features, 2 target classes) [35]. The records containing missing values were removed. All categorical variables in the datasets are converted to binary continuous values using one-hot encoding. For each dataset, 100 records are used for the explainer, 1000 records are used for testing, 1600 records are used for the auxiliary dataset, and the remaining records are split into 80% for the training dataset and 20% for the validation dataset. We change $|X_{\text{aux}}| = 100, 200, 400, 800, 1600$ for the experiments.

4.2 Implementations

The black-box model f to be explained was implemented using PyTorch [36] as a binary classification model with an input layer of dimension n , an output layer of dimension 2, and two hidden layers each containing $2n$ neurons. The activation function was softmax only in the output layer, and ReLU in all other layers. The attack model ψ was also implemented using PyTorch as a regression model with an input layer of dimension n , an output layer of dimension n , and one hidden layer containing $4n$ neurons. The activation function was sigmoid. Kernel SHAP [7] was used as an approximation method for calculating Shapley values. The Python library SHAP [7] provided by Lundberg et al. was used for implementation. The Python library LIME [8] provided by Ribeiro et al. was used for the implementation of LIME.

5 Results

5.1 Masked Rate and Record Reconstruction Risk

The results of MR and record reconstruction risk for SHAP are shown in Figs. 2 and 3, and the results for LIME are shown in Figs. 4 and 5. The error bars in the graphs represent 95% confidence intervals by calculating $\mu \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$. Regardless of the differences in XAI or datasets, as MR increases, the attacker’s MAE increases and SR decreases. Our proposed greedy privacy-aware masking achieves the strongest privacy protection at a given MR, yielding the largest increase in MAE and the steepest decrease in SR across datasets. In contrast, Global LeRF preserves most informative attribution entries and thus leads to the highest reconstruction risk, while Random and Global MoRF provide intermediate protection. Overall, the privacy risk ordering is consistent with the intuition that removing highly informative attribution coordinates (as prioritized by the greedy policy) most effectively degrades the attacker’s reconstruction performance.

5.2 Masked Rate and Infidelity

The results of MR and infidelity for SHAP and LIME are shown in Figs. 6, respectively. For SHAP, the infidelity curves are centered around the Random baseline: Global MoRF tends to yield higher infidelity, whereas Global LeRF tends to keep infidelity lower. In contrast, the proposed greedy privacy-aware masking shows a characteristic pattern: infidelity increases rapidly within the first 20% masking, and then remains nearly constant until the explanation is fully masked. For LIME,

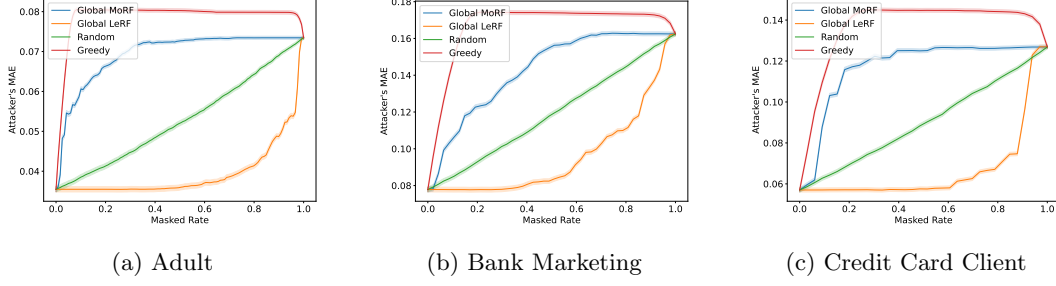


Fig. 2: Attacker's MAE when some parts of explanation vectors based on SHAP are masked

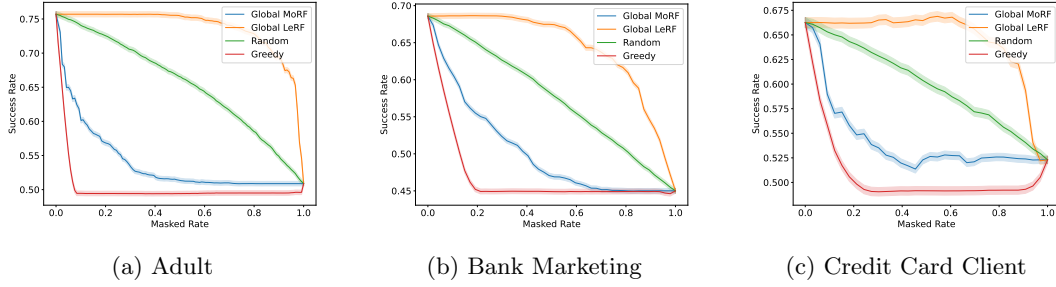


Fig. 3: SR when some parts of explanation vectors based on SHAP are masked

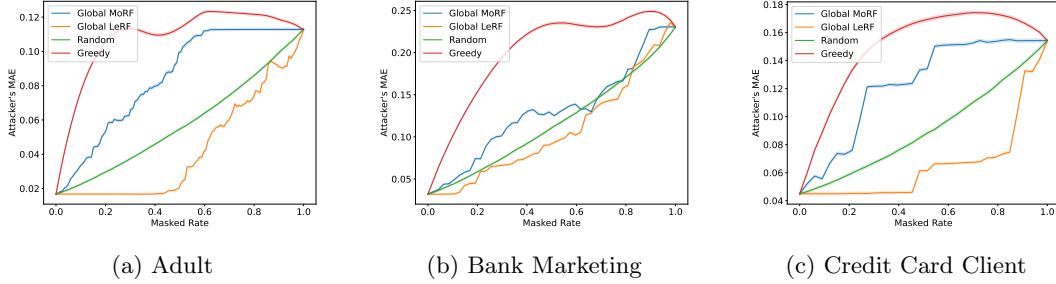


Fig. 4: Attacker's MAE when some parts of explanation vectors based on LIME are masked

the relationship between MR and infidelity is less stable, and the trends vary substantially across datasets. In particular, on the Adult dataset, masking can temporarily decrease infidelity, suggesting that removing some attribution coordinates may sometimes reduce noise in the explanation rather than degrade faithfulness.

5.3 Infidelity and Record Reconstruction Risk

The results of average infidelity and record reconstruction risk over three datasets and two XAI methods are shown in Figs. 7 and 8. When varying the MR, infidelity and reconstruction risk exhibit a much clearer monotonic relationship, implying that MR is a stronger driver of privacy degradation than infidelity itself. For SHAP, at the same infidelity level, the proposed greedy privacy-aware masking achieves the largest privacy improvement (higher MAE), while the other three baselines (Global MoRF/LeRF and Random) follow broadly similar trajectories. For LIME, the relationship is less consistent across datasets, which is plausibly due to the instability of infidelity for LIME under masking. Nevertheless, even for LIME, the greedy policy still tends to yield the lowest reconstruction risk among the compared masking strategies at matched infidelity.

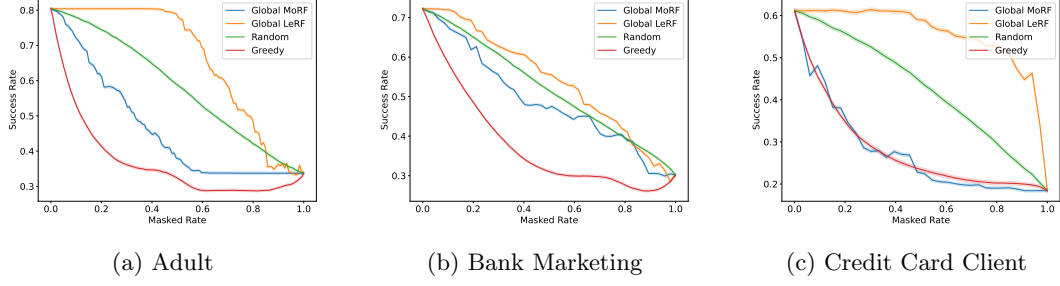


Fig. 5: SR when some parts of explanation vectors based on LIME are masked

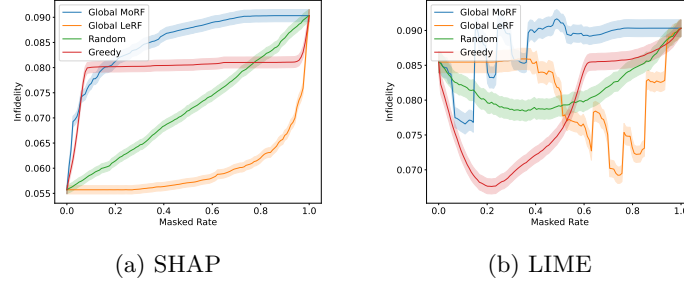


Fig. 6: Infidelity and Masked Rate MR of explanations for Adult dataset

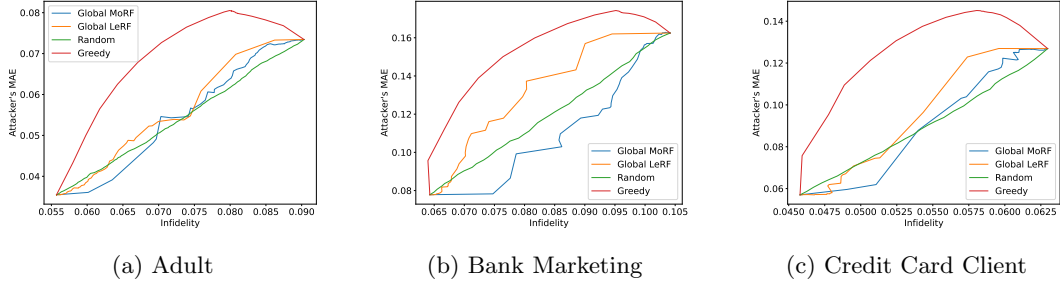


Fig. 7: Infidelity and Attacker's MAE w.r.t. MR of SHAP

5.4 Ablation Study

We conducted ablation study to confirm the effectiveness of each part in our proposal. We evaluate the AUCs (Area Under the Curves) for the attacker's MAE and infidelity by changing MR, shown in Table 1. The bold values mean the better result than the random baseline. According this result, our proposal shows the better MAE and almost same infidelity in comparison with the random. The proposal w/o privacy shows the best infidelity, however, attacker's MAE is worse than the random. On the other hand, the proposal w/o usefulness shows the best MAE while the infidelity is worst. Therefore, we can say that our proposal controls the privacy-utility trade-off and achieves the better performance than the baseline.

Table 1: Ablation study of AUCs in MR-MAE graph and MR-infidelity graph for four patterns: original, w/o privacy increase ΔP , w/o usefulness drop ΔU , and random baseline.

Masking Policies	AUC (MAE) $\uparrow$ (mean $\pm$ 95%CI)	AUC (infidelity) $\downarrow$ (mean $\pm$ 95%CI)
ours (original)	0.137 $\pm$ 0.007	0.072 $\pm$ 0.002
ours (w/o privacy)	0.071 $\pm$ 0.004	0.065 $\pm$ 0.003
ours (w/o usefulness)	0.154 $\pm$ 0.009	0.076 $\pm$ 0.003
random masking (baseline)	0.086 $\pm$ 0.005	0.071 $\pm$ 0.003

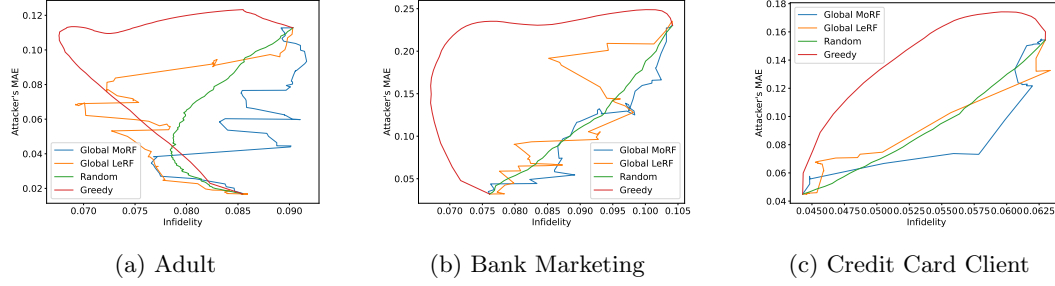


Fig. 8: Infidelity and Attacker's MAE w.r.t. MR of LIME

6 Limitations and Ethics

This study has several limitations: (1) experiments used only tabular data; we need to verify whether results transfer to image/text settings. (2) we tested only SHAP and LIME; other XAI methods (e.g., DeepLIFT [13], Grad-CAM [14]) should be evaluated. (3) We used only infidelity; the link between reconstruction risk and other metrics (e.g., F-Fidelity [31]) remains open. (4) The masking policy has notable overhead: full-test average runtime was 0.03,s (Global MoRF), 0.02,s (Global LeRF), 0.20,s (Random) vs 28.03,s for greedy privacy-aware masking (iterative per-record evaluation). Reducing this cost and addressing ethical issues are the key next steps. In healthcare, explanation masking can reduce inference of sensitive patient attributes; governance must define who receives explanations and at what masking level.

7 Conclusion

We investigated the relationship between record reconstruction risk and model explainability. We also proposed a masking method for explanations, Privacy-aware Masking, and experimentally evaluated its effectiveness. From the experimental results, we confirmed that masking policies can control the trade-off between privacy and utility. In addition, we showed that the masked rate reduces the record reconstruction risk, while the infidelity correlates weakly with the risk. For future work, it is important to investigate the relationship between the decrease in explainability due to privacy protection methods and the record reconstruction risk. In addition, it is necessary to investigate the relationship between other privacy risks and other explainability metrics.

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